

Crack-Extension Force for a Part-Through Crack in a Plate

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The crack stress-field parameter $\mathcal{K}$ and crack-extension force $\mathcal{G}$ at boundary points of a flat elliptical crack may be derived from knowledge that normal tension produces an ellipsoidal crack opening. Rough correction procedures can be employed to adapt this result for application to a part-through crack in a plate subjected to tension. Experimental measurements suggest this adapted result has a useful range of accuracy.

STARTING cracks which initiate crack-propagation failures usually develop from a local flaw of some kind which was considered unimportant or was not found by inspection. Sometimes the defect overlooked is an actual crack. At other times it is an arc strike, or a patch of material severely weakened by grain coarsening, or an excessively sharp root radius from which a crack may develop in service. These flaws as well as the highest tensile stress tend to occur at one border of the component cross section. In the examination of fracture failures one rarely finds that the initial starting crack extended completely through the thickness or that the starting-crack development, say by fatigue, occurred along a line extending completely through a cross section of a component. A part-through crack, as illustrated in Fig. 1, is more typical of the starting-crack shape customarily found.

In order to assist the use of laboratory measurements of crack toughness to estimate the danger of crack propagation under various conditions of tensile load and crack size it is desirable to estimate by calculation the stress-intensity factor $\mathcal{K}$ and force $\mathcal{G}$ descriptive of stress environment at the crack border. For a crack of the kind shown in Fig. 1 this is not an easy task even when the boundary is assumed to be representable as half an ellipse, a shape easily expressed in mathematical terms. The only three-dimensional crack problem for which stresses near the crack have been discussed in terms of values of $\mathcal{G}$ (or $\mathcal{K}$) is the crack with a circular boundary, sometimes referred to as a "penny-shaped" crack [1].¹

For a real crack similar to that illustrated in Fig. 1, the solution can be thought of as composed of (A) the solution to the mathematical problem of the flat elliptical crack in an infinite solid plus (B) consideration of three "real crack" factors as follows:

- 1 The correction for inserting a free surface normal to the crack through the major axis of the ellipse.
- 2 The correction for plastic strains near the crack boundary.
- 3 The correction for inserting a free surface representing the side of the plate opposite from the crack opening.

With the assistance of previous studies of the flat elliptical crack problem [2] it is possible to establish (A). This portion is discussed first. The discussion of the adjustments (B) which then follows is approximate only but is nevertheless desirable in order to clarify engineering use of the mathematical work.

¹ Numbers in brackets designate References at end of paper.

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The Basic Mathematical Model

Studies of the flat elliptical crack problem by Green and Sneddon [2] provide principally the information that tension normal to the crack produces an ellipsoidal crack-opening shape. Assume the crack lies in the x, z -plane with its major dimension $2c$ along the z -axis and its minor dimension $2a$ along the x -axis in such a way that border points x_1, z_1 of the crack correspond to

$$\frac{x_1^2}{a^2} + \frac{z_1^2}{c^2} = 1 \quad (1)$$

If the crack opening displacements in the y -direction are represented by η then the Green-Sneddon result may be expressed as

$$\eta = \eta_0 \left(1 - \frac{x^2}{a^2} - \frac{z^2}{c^2} \right)^{1/2} \quad (2)$$

where η_0 is half the total separation of the walls of the crack at the origin.

The desired expressions for $\mathcal{G}$ and $\mathcal{K}$ will be derived from equation (2) rather than from stress equations for two reasons: (a) The stress relations given by the Green-Sneddon paper are not in a form convenient for the calculation of σ_y near the border of the crack; (b) the author wishes to establish the point that the general shape of the crack opening provides sufficient information to determine $\mathcal{G}$ and $\mathcal{K}$.

The procedure to be followed consists first in a discussion of the variation around the border of the crack of the elastic-opening displacements of the crack. It will be noted that deviations of the stress state from plane strain become negligible in the limit of small separations from the crack border. Plane-strain relations for $\mathcal{G}$ and $\mathcal{K}$ associated with crack-opening displacement then permit the desired calculations.

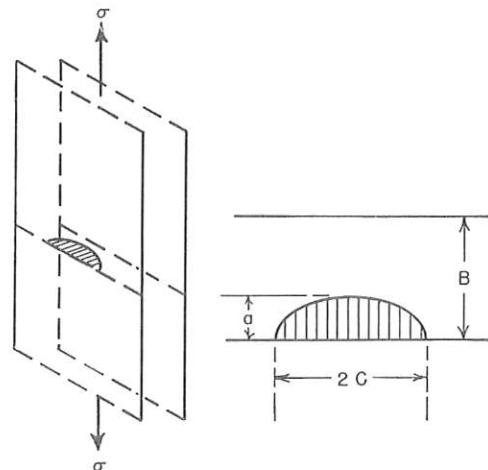


Fig. 1 Part-through crack in a plate showing dimensions a and c of crack and plate thickness, B

The position variables x_1 and z_1 which lie on the crack borders may be represented in parametric form by

$$x_1 = a \sin \phi \quad (3)$$

$$z_1 = c \cos \phi \quad (4)$$

A change $d\phi$ then corresponds to a segment ds of crack border given by

$$ds = (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} d\phi \quad (5)$$

At a small normal separation r (inward from the crack border) straightforward algebraic steps from equation (2) lead to the expression

$$\eta^2 = \eta_0^2 \frac{2r}{ac} (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} \quad (6)$$

The variation of displacement η at fixed distance r from the crack border as a function of ϕ will be considered in terms of the variation with ϕ of

$$\rho = \frac{\eta^2}{2r} \quad (7)$$

Thus

$$\rho = \frac{\eta_0^2}{ac} (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} \quad (8)$$

Differentiating with respect to ϕ and using equation (5) one finds

$$\frac{d\rho}{ds} = \left(\frac{\eta_0^2}{ac} \right) \frac{(c^2 - a^2) \sin \phi \cos \phi}{a^2 \cos^2 \phi + c^2 \sin^2 \phi} \quad (9)$$

At the value of ϕ where $d\rho/ds$ is largest the fractional change of ρ across the segment ds is

$$\frac{d\rho}{\rho} = \frac{(c^2 - a^2)(a^2 + c^2)^{1/2} ds}{2 \sqrt{2} a^2 c^2} \quad (10)$$

Putting $c = 2a$ for specific illustration, the value of $d\rho/\rho$ becomes

$$\frac{d\rho}{\rho} = 0.6 \frac{ds}{a} \quad (11)$$

The quantity ρ can be interpreted as the root radius of the elastic crack opening along the crack border. As a reference point approaches the crack border, ρ becomes the principal length factor associated with the crack. If we know a solution for stresses and strains valid along a crack-border length δs large enough so that the dimension ρ is negligible in comparison, then such a solution can be used to supply an expression for η for substitution into equation (6).

In crack stress-field analysis problems ρ is always equal to $(\sigma/E)^2$ times a dimension comparable to the crack size. Bearing in mind that linear elasticity analysis regards σ/E as an infinitesimal in comparison to unity, we can, for example, choose²

$$\delta s = \left(\frac{E}{\sigma} \right) \rho \quad (12)$$

$$\text{or, say,} \quad \delta s = \left(\frac{\sigma}{E} \right) a \quad (13)$$

Such a crack-border-length segment is obviously short compared to crack size and long compared to ρ . Substituting δs from equation (13) for ds in equation (11) gives

$$\frac{d\rho}{\rho} = 0.6 \left(\frac{\sigma}{E} \right) \quad (14)$$

which can be regarded as meaning ρ changes only by an in-

² E is Young's modulus, ν is Poisson's ratio.

finitesimal fraction across the border length segment δs . The changes of displacements and stresses near the crack border must possess a similar degree of constancy relative to a coordinate parallel to the crack border. Since the problem is symmetrical about the plane containing the crack, a plane-strain stress field fits the conditions of the problem in local crack-border regions comparable to dimensions to δs .

The three crack-border stress fields for which the displacements are independent of the co-ordinate parallel to the border are discussed in reference [3]. The first of these, opening-mode plane strain, is appropriate here and provides the relations

$$\eta = \frac{2(1 - \nu^2)}{E} (2r)^{1/2} \mathcal{K} \quad (15)$$

$$\mathcal{G} = \frac{\pi}{E} (1 - \nu^2) \mathcal{K}^2 \quad (16)$$

In reference [3] the $\mathcal{G}$ and $\mathcal{K}$ for opening-mode plane strain are written $\mathcal{G}_1$ and $\mathcal{K}_1$. The subscript is to distinguish these quantities from the $\mathcal{G}$ and $\mathcal{K}$ -values for the other two modes. Here the subscript can be temporarily omitted without risk of ambiguity. Inserting η^2 from equation (15) into equation (6) one finds

$$\mathcal{K}^2 = \frac{1}{4} \left(\frac{E}{1 - \nu^2} \right)^2 \left(\frac{\eta_0^2}{ac} \right) (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} \quad (17)$$

and thus

$$\mathcal{G} = \frac{\pi}{4} \left(\frac{E}{1 - \nu^2} \right) \left(\frac{\eta_0^2}{ac} \right) (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} \quad (18)$$

One can, at this point, either use a relationship of η_0 to σ , a , and c from reference [2] or find this relation by computing the strain-energy change for a small expansion of the crack boundary. The latter is not difficult. If the ellipse is expanded by adding fa to a and fc to c where f is very small, the normal outward displacement, r , of any point on the elliptical boundary is

$$r = \frac{acf}{(a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2}} \quad (19)$$

Computation of the strain-energy change dU from

$$dU = 4 \int_0^{\pi/2} \mathcal{G} r \frac{ds}{d\phi} d\phi \quad (20)$$

leads to

$$dU = \frac{\pi E c f \eta_0^2 \Phi}{(1 - \nu^2)} \quad (21)$$

where Φ is the elliptic integral

$$\Phi = \int_0^{\pi/2} \left[\sin^2 \phi + \left(\frac{a}{c} \right)^2 \cos^2 \phi \right]^{1/2} d\phi \quad (22)$$

We also know the strain-energy change is one half of σ times the change in crack volume. From this

$$dU = 2\pi \sigma a c f \eta_0 \quad (23)$$

The value of η_0 found by equating the two relations for dU , equations (21) and (23), may be substituted into (18) with the result that

$$\mathcal{G} = \frac{\pi(1 - \nu^2)\sigma^2}{E\Phi^2} \left(\frac{a}{c} \right) (a^2 \cos^2 \phi + c^2 \sin^2 \phi)^{1/2} \quad (24)$$

By inspection of equation (24) one observes that $\mathcal{G}$ is greatest where the crack boundary intersects the minor axis. Thus with increase of tension on a flat elliptical crack, the crack-extension should, barring anisotropy, tend to produce a circular crack-boundary shape.

The function Φ increases with (a/c) . When a is infinitesimal compared to c , Φ is unity. When the crack shape is circular Φ is $\pi/2$. Other values of Φ are easily found from published tables of elliptic integrals. Taking $\phi = \pi/2$ it is readily seen that the values of unity and $\pi/2$ for Φ correspond, respectively, to the Griffith equation

$$\mathcal{G} = \frac{\pi\sigma^2 a}{E} (1 - \nu^2) \quad (25)$$

and to the energy-release rate for the penny-shaped crack [1]

$$\mathcal{G} = \frac{4\sigma^2 a}{\pi E} (1 - \nu^2) \quad (26)$$

Approximate Estimates for Real Crack

Consider next the three corrections to this result which should be considered in constructing an estimate of $\mathcal{G}$ pertaining to a real part-through crack at one surface of a finite-thickness plate. No exact treatment of correction (2) is attempted. Although detailed numerical computations pertaining to corrections (1) and (3) might be somehow constructed, these would be overly elaborate considering that correction (2) can only be a rough approximation. In the subsequent discussion only estimates of a simple and approximate nature are employed. Some exercise of arbitrary choice will be noted in these approximations. Uncertainties remain in the final expression, equation (32), due to this. However, it is believed the treatment will serve to illuminate the nature of the "real crack" problem and that the end result has a useful degree of validity.

From general considerations it is clear that introduction of the two free surfaces allows larger crack-opening displacements. Corrections (1) and (3) therefore increase the crack-extension force. For reasons which will be discussed the influence of corrections (1) and (3) is assumed to be a 20 per cent increase of $\mathcal{G}$. Thus, for $\phi = \pi/2$,

$$\mathcal{G}_I = \frac{1.2\pi(1 - \nu^2)\sigma^2 a}{E\Phi^2} \quad (27)$$

When a is small both in relation to c and to the plate thickness, the factor 1.2 in the foregoing equation is known to represent correction (1) from the "crack at a free-surface boundary" solution discussed in reference [4]. Part-through cracks lie in a critical range of special interest when $a/c < 1.0$ and when $a/B \leq 0.5$, where B is the plate thickness as shown in Fig. 1. Within this range the small influence of the free surface opposite from the crack opening in causing increase of $\mathcal{G}$ should compensate somewhat for the overestimate of $\mathcal{G}$ by maintaining the 20 per cent correction as a/c increases. Thus it is tentatively assumed that equation (27) furnishes a useful estimate of the maximum $\mathcal{G}$ for (a/c) less than 1 and (a/B) less than 0.5. The fact that this estimate is not strictly limited to $\phi = \pi/2$ is discussed later.

The force-concept $\mathcal{G}$ employed in this paper pertains to a simple mathematical model of fracture consisting of a flat internal free surface, representing the crack, surrounded by a linear elastic-stress field. To avoid complexity and ambiguity in the force concept and yet make reasonable allowance for stress relaxation by plastic yielding near the crack edge, it is necessary to assume the influence of this local yielding on the surrounding stress field can be represented as equivalent to an additional spread of the internal free surface beyond the real crack size. It has been suggested in other papers [5, 6] that a suitable allowance for a situation of plane stress, where yielding is dominated by relaxation of the stress normal to a free surface, is an additional crack extension r_{YS} given by

$$r_{YS} = \frac{\mathcal{K}_I^2}{2\sigma_{YS}^2} \quad (28)$$

The term σ_{YS} represents the uniaxial tensile yield strength. When

yielding is limited by elastic constraint so that crack extension occurs in the transverse-tensile (plane-strain) mode a smaller value would be expected. The additional or equivalent crack extension suggested in reference [6] is

$$r_I = \frac{\mathcal{K}_I^2}{4\sqrt{2}\sigma_{YS}^2} \quad (29)$$

The subscript I implies the crack-edge stress field is governed by conditions which produce a situation of plane strain. With reference to this it can be noted that the lateral plate dimensions presumably are large compared to c . Thus plastic strains would not appreciably relax the stress σ_z close to the crack border and near $\phi = \pi/2$. Writing equation (27) with $\mathcal{K}_I$ rather than $\mathcal{G}_I$ terminology

$$\mathcal{K}_I^2 \Phi^2 = 1.2\sigma^2 a \quad (30)$$

Adding the amount given by equation (29) to a one obtains

$$\mathcal{K}_I^2 \Phi^2 = 1.2\sigma^2 \left(a + \frac{\mathcal{K}_I^2}{4\sqrt{2}\sigma_{YS}^2} \right) \quad (31)$$

and thus

$$\mathcal{K}_I^2 = \frac{1.2\sigma^2 a}{\Phi^2 - 0.212 \left(\frac{\sigma}{\sigma_{YS}} \right)^2} \quad (32)$$

The value of Φ depends only upon the ratio, a/c . There is no simple way of determining a plastic zone correction to the ratio a/c . Since the major axis lies on a free surface, one would guess the plastic zone at the extremities of the major axis would be larger than that at $\phi = \pi/2$, but this would depend upon the a/c ratio. Rather than attempt to estimate how much the effective a/c ratio should be shifted to larger or smaller size to allow for local plastic strain near the crack border, it will be assumed, for simplicity, that Φ can be calculated using the a/c ratio for the actual crack.

As a reference point moves on the crack border away from $\phi = \pi/2$ in either direction, variations of $\mathcal{K}_I^2$ in accordance with equation (24) might be expected. For $c = 2a$ the variation thus expected is less than 10 per cent across the central half of the z -direction span of the crack. This variation would be reduced by the tendency of the plastic-zone size to increase as the reference point approaches the intersection of the crack boundary with a free surface. Thus our estimate of $\mathcal{K}_I^2$ along a substantial portion of the crack boundary near $\phi = \pi/2$ should have nearly a constant value. It is suggested that this value can be calculated with useful accuracy from equation (32).

A résumé of the way the three corrections are taken into account by equation (32) is as follows: Correction (1) is assumed to increase $\mathcal{K}_I^2$ by 20 per cent. This correction factor is known to be valid when a/c is small. As a/c increases, the factor of increase representing correction (1) should decrease. However, increase of a/c usually corresponds to a closer approach of the crack boundary to the second free surface. Thus the moderate overestimate implied by maintaining the 20 per cent increase factor as the ratio a/c increases can be applied toward correction (3). Even a rough estimate of correction (3) would require extensive numerical computations. Since this discussion consistently avoids effects which cannot be handled in a simple way, it appears best to limit the magnitude of the second free-surface influence by limiting the a/B -values. In consideration of the fact that some allowance for correction (3) was implicit in the 20 per cent increase factor for correction (1), it is believed a range of a/B values extending to 0.5 would be permissible. Correction (2) was done by increasing the minor axis, a , an amount consistent with previous studies of plane-strain crack extension. The plastic-strain and stress-field conditions influencing the effective size of the major axis c are complex. For simplicity it was assumed the a/c ratio of the enlarged effective crack size would be the same as that for the actual crack.

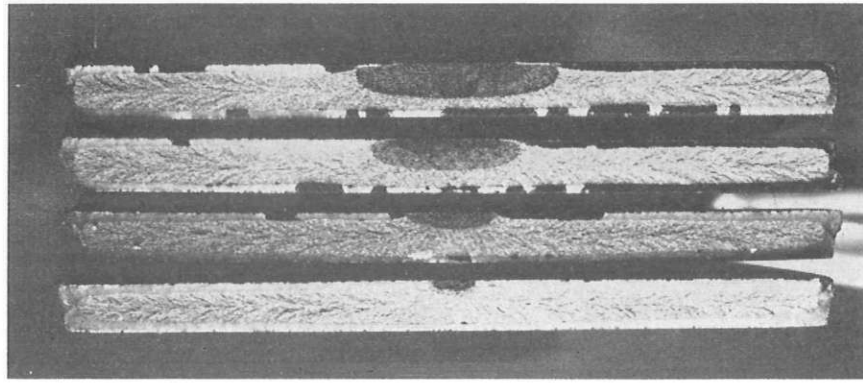


Fig. 2 Fracture appearance of tensile specimens which contained part-through cracks. Initial crack surface was darkened by furnace gases during heat-treatment which followed insertion of crack.

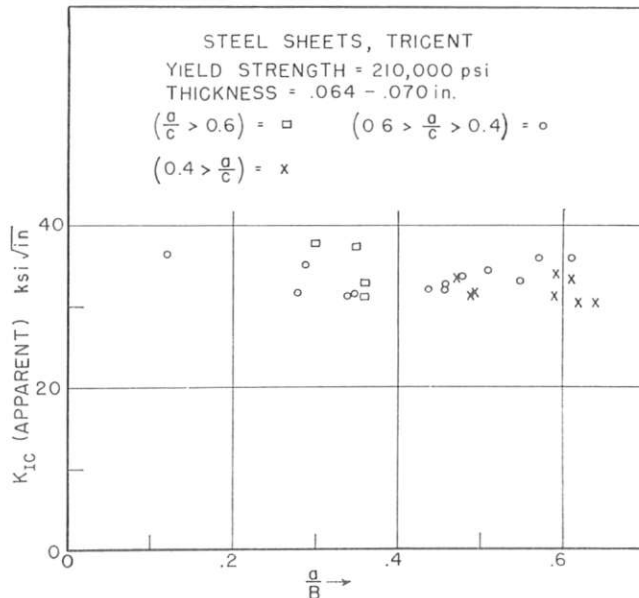


Fig. 3 Apparent values of $K_{Ic}(\sqrt{\pi} \mathcal{K}_{Ic})$ are shown as a function of relative crack depth for part-through cracks of different shapes as indicated by a/c ranges shown

Experimental Illustration

The tensile strengths of steel sheets containing various size part-through cracks have been measured by Srawley. The results appear in references [7] and [8]. After heat-treatment to produce a yield stress of 210,000 psi, the initial cracks were inserted by application of a corrosive agent to a limited region of the tensile side of the sheet while stressed in bending. Fig. 2 shows typical fracture surfaces from this work. The initial cracks were darkened by the furnace gases during a final baking for 16 hr at 400 F after cracking. From the tensile load at fracture and from measurements of the initial crack, values of σ , a , and c were obtained for use with equation (32).

References [7] and [8] give values K_{Ic} where

$$K_{Ic} = \sqrt{\pi} \mathcal{K}_{Ic} \quad (33)$$

Subscript c indicates that the computed crack edge-stress field parameter pertains to the stress field for onset of fast crack extension. There is some doubt as to this interpretation. The

results shown in Fig. 3 are therefore given as "apparent" K_{Ic} values.

The uncertainty stems from the fact that, although ink was used in the cracks to determine the amount of slow crack extension, the ink staining gave ambiguous indications and these were not usable for the purpose of establishing values of a and c based upon actual size of the crack at onset of rapid fracture. Customarily slow crack extensions marked by ink staining are comparable in magnitude to the size of the crack edge-plastic strain zone, less than 0.01 in. for the tests reported by Srawley. That some slow crack extension occurred and was assisted by the environmental effect of the ink can be inferred from a limited number of no-ink tests. These gave K_{Ic} results about 15 per cent above the average level of the results shown in Fig. 3.

It will be noted from the foregoing discussion that the uncertainties of the theoretical $\mathcal{K}_I$ estimate are scarcely greater than the uncertainties of experimental verification when one of the situations of practical interest, rocket vessel-wall material, is represented in a typical sheet thickness. For purposes of experimental verification of equation (32), $1/4$ -in.-thick plates and proportionately larger part-through cracks would have been preferable. Nevertheless, the results shown in Fig. 3 are encouraging with regard to the accuracy of equation (32). Within experimental scatter the relative depth and shape of the crack appear from these data to have no significant effect on the computed critical value of K_{Ic} .

References

- 1 I. N. Sneddon, "The Distribution of Stress in the Neighborhood of a Crack in an Elastic Solid," *Proceedings of the Physical Society of London*, vol. 187, 1946, p. 229.
- 2 A. E. Green and I. N. Sneddon, "The Distribution of Stress in the Neighborhood of a Flat Elliptical Crack in an Elastic Solid," *Proceedings of Cambridge Philosophical Society*, vol. 46, 1950, pp. 159-164.
- 3 G. R. Irwin, "Fracture Mechanics," *Structural Mechanics*, Pergamon Press, London, England, 1960, pp. 560-574.
- 4 G. R. Irwin, "Crack-Extension Force for a Crack at a Free Surface Boundary," *NRL Report 5120*, April, 1958.
- 5 "Fracture Testing of High-Strength Sheet Materials," First Report of a Special ASTM Committee, *ASTM Bulletin*, January, 1960, p. 38.
- 6 G. R. Irwin, "Plastic Zone Near a Crack and Fracture Toughness," 7th Sagamore Ordnance Materials Research Conference, August, 1960.
- 7 J. E. Srawley, "The Slow Growth and Rapid Propagation of Cracks," *NRL Report No. 5617*, May, 1961.
- 8 "The Slow Growth and Rapid Propagation of Cracks," Second Report of a Special ASTM Committee, *ASTM Materials Research and Standards*, vol. 1, no. 5, May, 1961.